LECTURE L23-24 · MODULE 4 · CO4

Transformation and scaling

Week 12 | Slides: `DATA209_Module_4_Feature_Engineering_and_Structure.pptx`, pages 3–11 | Laboratory: **P23-24**

TOPICS COVERED

- Log transform, Box–Cox
- Standardisation, MinMax, robust scaling
- Impact of transformation on the distribution

Why transform? Four reasons

1. **Reduce skew** — pull in a long tail so the bulk of the data occupies more of the plot and of

the statistic.

1. **Stabilise variance** — when spread grows with the mean, a log transform makes it roughly constant.
2. **Linearise** — a multiplicative relationship becomes additive under logs.
3. **Tame outliers** — compressing the tail reduces influence without deleting any row.

Transformation changes the units. Every interpretation afterwards is in transformed space — say so when you report.

The menu

TRANSFORM	EFFECT	REQUIRES
Log / log1p	strong pull-in of a right tail	strictly positive; log1p allows zero
Square root	milder than log	non-negative; good for counts
Reciprocal	very strong; reverses order	non-zero; rarely appropriate
Box–Cox	searches for the best power λ	strictly positive; λ = 0 is the log
Yeo–Johnson	Box–Cox generalised	handles zero and negatives
Rank / quantile	forces a chosen distribution	nothing — and is largely uninterpretable

Apply, then **re-measure**. A transform that does not reduce skew has not helped. Students routinely apply one and never check.

Transforming versus scaling

Transformation changes the *shape* (non-linear). **Scaling** changes *location and spread* (linear). Skew survives scaling completely untouched — so transform first if shape is the problem.

Order of operations: **impute → transform → scale**, and all of it after the split.

Four scalers

SCALER	METHOD	OUTPUT	BEST WHEN
StandardScaler	subtract mean, divide by SD	mean 0, SD 1	roughly symmetric; PCA, linear models
MinMaxScaler	rescale to a fixed interval	usually 0 to 1	bounded input needed, no extreme outliers
RobustScaler	subtract median, divide by IQR	unbounded	outliers present and being kept
MaxAbsScaler	divide by largest absolute value	−1 to 1	sparse data; preserves zeros

One extreme value ruins MinMaxScaler: the range stretches and every ordinary value compresses toward zero.

Does the model care?

METHOD	NEEDS SCALING	WHY
k-NN, k-means	yes	distance-based; a large-range variable dominates
PCA	yes	maximises variance — unscaled, one variable takes PC1 alone
SVM, neural networks	yes	convergence and regularisation depend on scale
Ridge, Lasso	yes	the penalty applies to coefficients, whose size depends on scale
Linear regression (unpenalised)	no	coefficients absorb scale
Trees, random forest, boosting	no	splits are threshold-based, invariant to monotone rescaling

When in doubt, scale. The cost is one pipeline step.

KEY TERMS

`Log transform` `Box–Cox` `Yeo–Johnson` `Variance stabilisation` `StandardScaler` `MinMaxScaler` `RobustScaler` `MaxAbsScaler` `Scale invariance`

COMMON ERRORS TO AVOID

- Scaling a skewed variable and expecting the skew to change
- Using MinMaxScaler on data with extreme values
- Applying Box–Cox to a column containing zeros
- Not re-measuring skew after transforming

READINGS — ALL FREE TO ACCESS

1. scikit-learn — comparing the effect of different scalers
   `https://scikit-learn.org/stable/auto_examples/preprocessing/plot_all_scaling.html`
2. scikit-learn — preprocessing
   `https://scikit-learn.org/stable/modules/preprocessing.html`
3. Kuhn & Johnson — Feature Engineering and Selection
   `http://www.feat.engineering/`

TEST YOURSELF

1. A column has skew 2.3 and contains zeros. Which transform, and why not Box–Cox?
2. Why does scaling leave skewness unchanged?
3. Which of random forest and k-NN needs scaled input, and why?
4. Name the correct order of impute, scale, transform and split.